Data appendix — benchmax

Every table below is generated from `results/raw/runs.jsonl` by `python -m benchmax.cli report`. Figures are in `reports/figures/`. n = 3 replicates per (backend, model, quantization, workload) unless noted.

Environment

key	value
timestamp_utc	2026-10-15T02:42:07Z
platform	Linux-6.6.87.2-microsoft-standard-WSL2-x86_64-with-glibc2.39
python	3.12.3
logical_cores	20
gpu	NVIDIA GeForce RTX 4060 Laptop GPU, 8188 MiB, 610.62, 8.9
cpu	Intel(R) Core(TM) i7-14650HX
ram_gib	10.69
wsl_note	running inside WSL2; visible RAM is the .wslconfig cap, which is smaller than host RAM

Backend versions

key	value
llamacpp	{'commit': 'a0e6dc7384edb8f5d927155e1d26c10b4a310e6d', 'build': 'GGML_CUDA=ON, CMAKE_CUDA_ARCHITECTURES=89'}
vllm	0.31.0

Matrix (every cell and why it was or was not run)

backend	model	quant	scheme	status	reason
llamacpp	qwen3-0.6b	4bit	Q4_K_M (GGUF)	ready	
llamacpp	qwen3-0.6b	8bit	Q8_0 (GGUF)	ready	
vllm	qwen3-0.6b	8bit	FP8 (W8A8, e4m3)	ready	

Performance rollup (all workloads combined)

Latency in ms, throughput in tok/s, memory in MiB. Averaged across all four workloads, so use the per-workload table for concurrency-specific claims.

backend	model	quant_level	ttft_ms	ttft_sd	tpot_ms	itl_p99	tok_s	goodput	vram_mib	n_reps
llamacpp	Llama-3.2-1B-Instruct	16bit	993.82	1,564.1	13.67	15.66	268.61	183.61	4637	12
llamacpp	Llama-3.2-1B-Instruct	4bit	922.71	1,520.9	10.02	13.91	337.27	242.46	3019	12
llamacpp	Llama-3.2-1B-Instruct	8bit	929.87	1,585.8	11.29	16.65	314.28	230.67	3532	12
llamacpp	Llama-3.2-3B-Instruct	4bit	2,343.2	3,278.5	19.97	25.99	152.24	82.75	5270	12
llamacpp	Llama-3.2-3B-Instruct	8bit	1,849.2	2,729.8	19.76	21.10	164.97	76.05	6733	12
llamacpp	Qwen3-0.6B	16bit	978.81	1,632.9	10.29	14.79	332.87	246.13	3699	12
llamacpp	Qwen3-0.6B	4bit	882.35	1,443.1	8.99	12.22	370.46	273.68	2994	12
llamacpp	Qwen3-0.6B	8bit	940.14	1,529.2	9.42	13.72	349.35	257.03	3481	12
llamacpp	Qwen3-1.7B	16bit	2,080.7	2,834.3	20.07	24.09	154.74	71.91	5980	12
llamacpp	Qwen3-1.7B	4bit	1,117.0	1,835.5	13.04	14.61	261.20	189.05	3832	12
llamacpp	Qwen3-1.7B	8bit	1,206.0	1,964.8	14.55	17.45	237.29	166.30	4507	12
llamacpp	Qwen3-4B	4bit	2,327.2	3,841.7	26.84	30.36	127.81	92.86	5951	12
llamacpp	Qwen3-4B	8bit	2,478.7	3,599.6	26.98	31.52	121.43	56.31	7568	12
vllm	Llama-3.2-1B-Instruct	16bit	701.22	1,119.0	70.76	127.57	126.30	94.70	7331	12
vllm	Llama-3.2-1B-Instruct	8bit	449.94	743.54	48.11	76.26	172.52	139.88	7218	12
vllm	Llama-3.2-3B-Instruct	8bit	590.35	823.61	83.93	176.62	109.69	48.57	7389	12
vllm	Qwen3-0.6B	16bit	49.46	40.12	8.46	10.88	1,015.0	1,015.0	6981	12
vllm	Qwen3-0.6B	8bit	46.26	45.28	6.44	8.91	1,220.3	1,220.3	7240	12
vllm	Qwen3-0.6B	int8	46.00	46.22	6.59	9.25	1,139.9	1,139.9	7426	12
vllm	Qwen3-1.7B	16bit	88.10	106.99	17.28	21.16	509.97	509.97	6882	12
vllm	Qwen3-1.7B	8bit	90.71	129.68	12.16	15.30	682.09	682.09	6970	12
vllm	Qwen3-1.7B	int8	88.30	121.12	12.15	15.82	682.13	682.13	7067	12
vllm	Qwen3-4B	4bit	179.12	293.80	16.44	19.89	476.09	440.61	7150	12
vllm	Qwen3-4B	8bit	190.08	318.25	22.63	32.14	372.14	337.84	7415	12

Performance by workload, backend and quantization

This is the table to cite for any concurrency-specific claim.

workload	backend	quant_level	ttft_ms	ttft_sd	ttft_p95	tpot_ms	tpot_sd	tpot_p95	itl_p50	itl_p9
w1_single_short	llamacpp	16bit	34.06	43.63	34.06	11.38	3.53	11.38	11.28	12.57
w1_single_short	llamacpp	4bit	21.98	11.85	21.98	7.82	3.50	7.82	7.64	9.62
w1_single_short	llamacpp	8bit	25.89	12.44	25.89	11.24	5.86	11.24	11.19	12.74
w1_single_short	vllm	16bit	42.34	32.27	42.34	29.55	27.08	29.55	29.95	39.20
w1_single_short	vllm	4bit	28.81	7.23	28.81	13.86	0.37	13.86	13.74	17.19
w1_single_short	vllm	8bit	48.20	38.05	48.20	31.90	31.15	31.90	32.07	41.91
w1_single_short	vllm	int8	18.25	4.38	18.25	8.07	2.95	8.07	7.99	10.48
w2_single_prefill	llamacpp	16bit	53.76	63.69	53.76	12.20	4.05	12.20	12.13	13.67
w2_single_prefill	llamacpp	4bit	73.49	109.39	73.49	8.38	3.80	8.38	8.30	10.12
w2_single_prefill	llamacpp	8bit	67.73	92.12	67.73	11.93	6.17	11.93	11.85	13.32
w2_single_prefill	vllm	16bit	206.18	431.62	206.18	30.48	27.79	30.48	30.68	36.31
w2_single_prefill	vllm	4bit	123.28	163.21	123.28	14.42	0.33	14.42	14.33	17.06
w2_single_prefill	vllm	8bit	207.63	354.36	207.63	33.79	30.80	33.79	31.95	47.71
w2_single_prefill	vllm	int8	44.59	46.69	44.59	8.23	3.05	8.23	8.19	10.64
w3_concurrent_8	llamacpp	16bit	747.74	1,082.5	751.33	16.78	5.27	16.78	16.56	21.09
w3_concurrent_8	llamacpp	4bit	587.34	856.16	593.53	23.21	9.98	23.22	22.96	29.41
w3_concurrent_8	llamacpp	8bit	655.70	825.94	660.49	20.58	7.14	20.61	20.29	26.04
w3_concurrent_8	vllm	16bit	253.60	501.35	282.05	32.25	31.37	33.54	30.77	63.18
w3_concurrent_8	vllm	4bit	129.71	106.40	142.07	14.76	0.13	15.21	14.53	18.65
w3_concurrent_8	vllm	8bit	230.66	347.11	254.24	31.73	25.50	32.67	29.50	69.58
w3_concurrent_8	vllm	int8	50.87	29.92	54.76	8.58	2.79	8.68	8.51	10.76
w4_concurrent_32	llamacpp	16bit	4,568.9	1,480.7	8,933.7	18.34	4.59	20.26	17.65	25.39
w4_concurrent_32	llamacpp	4bit	5,391.2	2,386.4	10,600.0	23.68	10.69	26.08	22.96	28.53
w4_concurrent_32	llamacpp	8bit	5,173.8	1,996.8	10,208.6	21.85	7.41	24.63	21.01	28.24
w4_concurrent_32	vllm	16bit	616.25	1,226.5	800.14	36.38	31.66	37.37	31.11	74.14
w4_concurrent_32	vllm	4bit	434.68	544.55	587.73	22.72	1.59	24.95	21.75	26.64
w4_concurrent_32	vllm	8bit	607.38	908.53	835.35	41.19	33.34	42.96	35.43	88.18
w4_concurrent_32	vllm	int8	154.89	151.62	199.38	12.61	3.44	12.90	12.34	18.28

TTFT / throughput frontier

`on_frontier = True` means no other configuration beats this one on both throughput and TTFT.

workload	backend	model	quant_level	ttft_ms	out_tok_s	tpot_ms	goodput_tok_s	on_frontier
w1_single_short	llamacpp	Llama-3.2-1B-Instruct	16bit	18.47	88.21	11.27	88.21	True
w1_single_short	llamacpp	Llama-3.2-1B-Instruct	4bit	13.33	194.72	5.07	194.72	True
w1_single_short	llamacpp	Llama-3.2-1B-Instruct	8bit	17.82	133.71	7.39	133.71	True
w1_single_short	llamacpp	Llama-3.2-3B-Instruct	4bit	25.20	98.22	10.06	98.22	False
w1_single_short	llamacpp	Llama-3.2-3B-Instruct	8bit	30.65	66.89	14.82	66.89	False
w1_single_short	llamacpp	Qwen3-0.6B	16bit	15.35	134.69	7.36	134.69	True
w1_single_short	llamacpp	Qwen3-0.6B	4bit	21.77	238.84	4.05	238.84	False
w1_single_short	llamacpp	Qwen3-0.6B	8bit	19.61	214.71	4.54	214.71	False
w1_single_short	llamacpp	Qwen3-1.7B	16bit	68.34	62.83	15.51	62.83	False
w1_single_short	llamacpp	Qwen3-1.7B	4bit	20.11	146.11	6.74	146.11	False
w1_single_short	llamacpp	Qwen3-1.7B	8bit	20.95	108.67	9.11	108.67	False
w1_single_short	llamacpp	Qwen3-4B	4bit	29.50	75.00	13.20	75.00	False
w1_single_short	llamacpp	Qwen3-4B	8bit	40.44	48.75	20.35	48.75	True
w1_single_short	vllm	Llama-3.2-1B-Instruct	16bit	82.17	15.29	65.31	15.29	True
w1_single_short	vllm	Llama-3.2-1B-Instruct	8bit	54.40	31.46	36.15	31.46	True
w1_single_short	vllm	Llama-3.2-3B-Instruct	8bit	116.65	11.42	87.40	11.42	True
w1_single_short	vllm	Qwen3-0.6B	16bit	19.77	130.22	7.59	130.22	False
w1_single_short	vllm	Qwen3-0.6B	8bit	17.40	180.85	5.46	180.85	False
w1_single_short	vllm	Qwen3-0.6B	int8	14.41	183.51	5.38	183.51	True
w1_single_short	vllm	Qwen3-1.7B	16bit	25.07	63.23	15.74	63.23	True
w1_single_short	vllm	Qwen3-1.7B	8bit	23.40	92.35	10.73	92.35	False
w1_single_short	vllm	Qwen3-1.7B	int8	22.09	92.19	10.76	92.19	False
w1_single_short	vllm	Qwen3-4B	4bit	28.81	71.55	13.86	71.55	False
w1_single_short	vllm	Qwen3-4B	8bit	29.17	50.37	19.78	50.37	True
w2_single_prefill	llamacpp	Llama-3.2-1B-Instruct	16bit	40.89	83.95	11.46	83.95	True
w2_single_prefill	llamacpp	Llama-3.2-1B-Instruct	4bit	33.45	185.74	5.01	185.74	False
w2_single_prefill	llamacpp	Llama-3.2-1B-Instruct	8bit	37.85	128.62	7.33	128.62	True
w2_single_prefill	llamacpp	Llama-3.2-3B-Instruct	4bit	102.84	86.38	10.43	86.38	False
w2_single_prefill	llamacpp		8bit	83.78	60.91	15.45	60.91	False

workload	backend	model	quant_level	ttft_ms	out_tok_s	tpot_ms	goodput_tok_s	on_frontier
		Llama-3.2-3B-Instruct						
w2_single_prefill	llamacpp	Qwen3-0.6B	16bit	39.76	119.04	7.95	119.04	True
w2_single_prefill	llamacpp	Qwen3-0.6B	4bit	42.28	192.50	4.68	192.50	False
w2_single_prefill	llamacpp	Qwen3-0.6B	8bit	34.42	185.57	4.97	185.57	False
w2_single_prefill	llamacpp	Qwen3-1.7B	16bit	80.64	55.18	17.20	55.18	True
w2_single_prefill	llamacpp	Qwen3-1.7B	4bit	56.06	125.30	7.32	125.30	False
w2_single_prefill	llamacpp	Qwen3-1.7B	8bit	57.05	90.74	10.34	90.74	False
w2_single_prefill	llamacpp	Qwen3-4B	4bit	132.82	62.62	14.46	62.62	False
w2_single_prefill	llamacpp	Qwen3-4B	8bit	125.54	43.38	21.57	43.38	True
w2_single_prefill	vllm	Llama-3.2-1B-Instruct	16bit	516.09	13.71	67.23	9.96	False
w2_single_prefill	vllm	Llama-3.2-1B-Instruct	8bit	343.96	19.19	48.47	19.19	True
w2_single_prefill	vllm	Llama-3.2-3B-Instruct	8bit	477.76	11.29	83.21	7.01	True
w2_single_prefill	vllm	Qwen3-0.6B	16bit	39.92	116.59	8.09	116.59	True
w2_single_prefill	vllm	Qwen3-0.6B	8bit	28.96	175.94	5.32	175.94	False
w2_single_prefill	vllm	Qwen3-0.6B	int8	28.07	172.33	5.45	172.33	True
w2_single_prefill	vllm	Qwen3-1.7B	16bit	62.53	59.42	16.13	59.42	True
w2_single_prefill	vllm	Qwen3-1.7B	8bit	56.84	86.54	10.87	86.54	False
w2_single_prefill	vllm	Qwen3-1.7B	int8	61.11	85.12	11.02	85.12	False
w2_single_prefill	vllm	Qwen3-4B	4bit	123.28	62.87	14.42	62.87	False
w2_single_prefill	vllm	Qwen3-4B	8bit	130.61	44.18	21.08	44.18	False
w3_concurrent_8	llamacpp	Llama-3.2-1B-Instruct	16bit	361.75	448.95	15.30	448.95	False
w3_concurrent_8	llamacpp	Llama-3.2-1B-Instruct	4bit	269.70	462.95	15.50	462.95	False
w3_concurrent_8	llamacpp	Llama-3.2-1B-Instruct	8bit	129.30	548.88	13.64	548.88	False
w3_concurrent_8	llamacpp	Llama-3.2-3B-Instruct	4bit	1,758.2	199.81	28.73	90.27	False
w3_concurrent_8	llamacpp	Llama-3.2-3B-Instruct	8bit	1,044.6	262.53	23.31	109.02	False
w3_concurrent_8	llamacpp	Qwen3-0.6B	16bit	185.63	615.11	11.69	615.11	False
w3_concurrent_8	llamacpp	Qwen3-0.6B	4bit	202.30	534.34	13.50	534.34	False
w3_concurrent_8	llamacpp	Qwen3-0.6B	8bit	240.80	504.71	14.10	504.71	False
w3_concurrent_8	llamacpp	Qwen3-1.7B	16bit	1,695.8	233.15	23.34	102.67	False
w3_concurrent_8	llamacpp	Qwen3-1.7B	4bit	239.91	388.60	18.94	388.60	False
w3_concurrent_8	llamacpp	Qwen3-1.7B	8bit	299.88	371.16	19.61	371.16	False

workload	backend	model	quant_level	ttft_ms	out_tok_s	tpot_ms	goodput_tok_s	on_frontier
w3_concurrent_8	llamacpp	Qwen3-4B	4bit	466.59	187.22	39.38	187.22	False
w3_concurrent_8	llamacpp	Qwen3-4B	8bit	1,564.0	188.36	32.23	81.80	False
w3_concurrent_8	vllm	Llama-3.2-1B-Instruct	16bit	652.55	105.55	72.70	75.85	True
w3_concurrent_8	vllm	Llama-3.2-1B-Instruct	8bit	350.88	153.51	49.68	153.51	True
w3_concurrent_8	vllm	Llama-3.2-3B-Instruct	8bit	570.45	107.49	70.59	78.17	True
w3_concurrent_8	vllm	Qwen3-0.6B	16bit	38.87	1,024.5	7.52	1,024.5	True
w3_concurrent_8	vllm	Qwen3-0.6B	8bit	38.30	1,302.3	5.85	1,302.3	False
w3_concurrent_8	vllm	Qwen3-0.6B	int8	38.15	1,266.3	6.04	1,266.3	True
w3_concurrent_8	vllm	Qwen3-1.7B	16bit	69.38	470.97	16.53	470.97	True
w3_concurrent_8	vllm	Qwen3-1.7B	8bit	64.26	679.11	11.33	679.11	True
w3_concurrent_8	vllm	Qwen3-1.7B	int8	63.59	691.55	11.12	691.55	True
w3_concurrent_8	vllm	Qwen3-4B	4bit	129.71	509.56	14.76	509.56	False
w3_concurrent_8	vllm	Qwen3-4B	8bit	129.41	362.52	21.20	362.52	True
w4_concurrent_32	llamacpp	Llama-3.2-1B-Instruct	16bit	3,554.2	453.32	16.65	113.33	False
w4_concurrent_32	llamacpp	Llama-3.2-1B-Instruct	4bit	3,374.4	505.68	14.49	126.42	False
w4_concurrent_32	llamacpp	Llama-3.2-1B-Instruct	8bit	3,534.5	445.91	16.80	111.48	False
w4_concurrent_32	llamacpp	Llama-3.2-3B-Instruct	4bit	7,486.7	224.56	30.66	56.14	True
w4_concurrent_32	llamacpp	Llama-3.2-3B-Instruct	8bit	6,237.8	269.55	25.44	67.39	True
w4_concurrent_32	llamacpp	Qwen3-0.6B	16bit	3,674.5	462.63	14.16	115.66	False
w4_concurrent_32	llamacpp	Qwen3-0.6B	4bit	3,263.0	516.16	13.74	129.04	False
w4_concurrent_32	llamacpp	Qwen3-0.6B	8bit	3,465.8	492.42	14.08	123.10	False
w4_concurrent_32	llamacpp	Qwen3-1.7B	16bit	6,478.1	267.80	24.22	66.95	True
w4_concurrent_32	llamacpp	Qwen3-1.7B	4bit	4,152.1	384.77	19.17	96.19	False
w4_concurrent_32	llamacpp	Qwen3-1.7B	8bit	4,446.2	378.57	19.15	94.64	False
w4_concurrent_32	llamacpp	Qwen3-4B	4bit	8,680.0	186.38	40.34	46.60	True
w4_concurrent_32	llamacpp	Qwen3-4B	8bit	8,185.0	205.25	33.77	51.31	True
w4_concurrent_32	vllm	Llama-3.2-1B-Instruct	16bit	1,554.1	370.65	77.79	277.70	False
w4_concurrent_32	vllm	Llama-3.2-1B-Instruct	8bit	1,050.5	485.94	58.13	355.35	True
w4_concurrent_32	vllm	Llama-3.2-3B-Instruct	8bit	1,196.5	308.59	94.53	97.68	True

workload	backend	model	quant_level	ttft_ms	out_tok_s	tpot_ms	goodput_tok_s	on_frontier
w4_concurrent_32	vllm	Qwen3-0.6B	16bit	99.27	2,788.7	10.66	2,788.7	True
w4_concurrent_32	vllm	Qwen3-0.6B	8bit	100.36	3,222.3	9.11	3,222.3	False
w4_concurrent_32	vllm	Qwen3-0.6B	int8	103.37	2,937.3	9.50	2,937.3	False
w4_concurrent_32	vllm	Qwen3-1.7B	16bit	195.43	1,446.3	20.70	1,446.3	True
w4_concurrent_32	vllm	Qwen3-1.7B	8bit	218.33	1,870.4	15.70	1,870.4	False
w4_concurrent_32	vllm	Qwen3-1.7B	int8	206.40	1,859.7	15.72	1,859.7	False
w4_concurrent_32	vllm	Qwen3-4B	4bit	434.68	1,260.4	22.72	1,118.5	True
w4_concurrent_32	vllm	Qwen3-4B	8bit	471.13	1,031.5	28.48	894.31	True

Quantization cost — perplexity

Normalised per model against that model's own unquantized run. Cross-backend perplexity is NOT comparable (different context windows and tools).

axis	backend	model	quant_level	quant_scheme	value	reference	ratio	workload
perplexity	llamacpp	Llama-3.2-1B-Instruct	4bit	Q4_K_M (GGUF)	10.64	10.42	1.02	
perplexity	llamacpp	Llama-3.2-1B-Instruct	8bit	Q8_0 (GGUF)	10.41	10.42	1.00	
perplexity	llamacpp	Llama-3.2-1B-Instruct	16bit	F16 (GGUF)	10.42	10.42	1.00	
perplexity	llamacpp	Qwen3-0.6B	16bit	BF16 (GGUF)	15.25	15.25	1.00	
perplexity	llamacpp	Qwen3-1.7B	4bit	Q4_K_M (GGUF)	13.37	13.04	1.03	
perplexity	llamacpp	Qwen3-1.7B	8bit	Q8_0 (GGUF)	13.00	13.04	1.00	
perplexity	llamacpp	Qwen3-1.7B	16bit	BF16 (GGUF)	13.04	13.04	1.00	
perplexity	vllm	Llama-3.2-1B-Instruct	8bit	FP8 dynamic (W8A8)	10.74	10.67	1.01	
perplexity	vllm	Llama-3.2-1B-Instruct	16bit	BF16 (unquantized)	10.67	10.67	1.00	
perplexity	vllm	Qwen3-0.6B	int8	GPTQ-INT8 (W8A16)	15.00	14.94	1.00	
perplexity	vllm	Qwen3-0.6B	8bit	FP8 (W8A8, e4m3)	15.22	14.94	1.02	
perplexity	vllm	Qwen3-0.6B	16bit	BF16 (unquantized)	14.94	14.94	1.00	
perplexity	vllm	Qwen3-1.7B	int8	GPTQ-INT8 (W8A16)	12.52	12.52	1.00	
perplexity	vllm	Qwen3-1.7B	8bit	FP8 (W8A8, e4m3)	12.43	12.52	0.99	
perplexity	vllm	Qwen3-1.7B	16bit	BF16 (unquantized)	12.52	12.52	1.00	

Quantization cost — throughput

Normalised per (backend, model) against the same configuration's 16-bit reference.

axis	backend	model	quant_level	quant_scheme	value	reference	ratio	workload
throughput	llamacpp	llama32-1b	16bit	F16 (GGUF)	88.21	88.21	1.00	w1_single_short
throughput	llamacpp	llama32-1b	4bit	Q4_K_M (GGUF)	194.72	88.21	2.21	w1_single_short
throughput	llamacpp	llama32-1b	8bit	Q8_0 (GGUF)	133.71	88.21	1.52	w1_single_short
throughput	llamacpp	qwen3-0.6b	16bit	BF16 (GGUF)	134.69	134.69	1.00	w1_single_short
throughput	llamacpp	qwen3-0.6b	4bit	Q4_K_M (GGUF)	238.84	134.69	1.77	w1_single_short
throughput	llamacpp	qwen3-0.6b	8bit	Q8_0 (GGUF)	214.71	134.69	1.59	w1_single_short
throughput	llamacpp	qwen3-1.7b	16bit	BF16 (GGUF)	62.83	62.83	1.00	w1_single_short
throughput	llamacpp	qwen3-1.7b	4bit	Q4_K_M (GGUF)	146.11	62.83	2.33	w1_single_short
throughput	llamacpp	qwen3-1.7b	8bit	Q8_0 (GGUF)	108.67	62.83	1.73	w1_single_short
throughput	vllm	llama32-1b	16bit	BF16 (unquantized)	15.29	15.29	1.00	w1_single_short
throughput	vllm	llama32-1b	8bit	FP8 dynamic (W8A8)	31.46	15.29	2.06	w1_single_short
throughput	vllm	qwen3-0.6b	16bit	BF16 (unquantized)	130.22	130.22	1.00	w1_single_short
throughput	vllm	qwen3-0.6b	8bit	FP8 (W8A8, e4m3)	180.85	130.22	1.39	w1_single_short
throughput	vllm	qwen3-0.6b	int8	GPTQ-INT8 (W8A16)	183.51	130.22	1.41	w1_single_short
throughput	vllm	qwen3-1.7b	16bit	BF16 (unquantized)	63.23	63.23	1.00	w1_single_short
throughput	vllm	qwen3-1.7b	8bit	FP8 (W8A8, e4m3)	92.35	63.23	1.46	w1_single_short
throughput	vllm	qwen3-1.7b	int8	GPTQ-INT8 (W8A16)	92.19	63.23	1.46	w1_single_short
throughput	llamacpp	llama32-1b	16bit	F16 (GGUF)	83.95	83.95	1.00	w2_single_prefill
throughput	llamacpp	llama32-1b	4bit	Q4_K_M (GGUF)	185.74	83.95	2.21	w2_single_prefill
throughput	llamacpp	llama32-1b	8bit	Q8_0 (GGUF)	128.62	83.95	1.53	w2_single_prefill
throughput	llamacpp	qwen3-0.6b	16bit	BF16 (GGUF)	119.04	119.04	1.00	w2_single_prefill
throughput	llamacpp	qwen3-0.6b	4bit	Q4_K_M (GGUF)	192.50	119.04	1.62	w2_single_prefill
throughput	llamacpp	qwen3-0.6b	8bit	Q8_0 (GGUF)	185.57	119.04	1.56	w2_single_prefill
throughput	llamacpp	qwen3-1.7b	16bit	BF16 (GGUF)	55.18	55.18	1.00	w2_single_prefill
throughput	llamacpp	qwen3-1.7b	4bit	Q4_K_M (GGUF)	125.30	55.18	2.27	w2_single_prefill
throughput	llamacpp	qwen3-1.7b	8bit	Q8_0 (GGUF)	90.74	55.18	1.64	w2_single_prefill
throughput	vllm	llama32-1b	16bit	BF16 (unquantized)	13.71	13.71	1.00	w2_single_prefill
throughput	vllm	llama32-1b	8bit	FP8 dynamic (W8A8)	19.19	13.71	1.40	w2_single_prefill
throughput	vllm	qwen3-0.6b	16bit	BF16 (unquantized)	116.59	116.59	1.00	w2_single_prefill

axis	backend	model	quant_level	quant_scheme	value	reference	ratio	workload
throughput	vllm	qwen3-0.6b	8bit	FP8 (W8A8, e4m3)	175.94	116.59	1.51	w2_single_prefill
throughput	vllm	qwen3-0.6b	int8	GPTQ-INT8 (W8A16)	172.33	116.59	1.48	w2_single_prefill
throughput	vllm	qwen3-1.7b	16bit	BF16 (unquantized)	59.42	59.42	1.00	w2_single_prefill
throughput	vllm	qwen3-1.7b	8bit	FP8 (W8A8, e4m3)	86.54	59.42	1.46	w2_single_prefill
throughput	vllm	qwen3-1.7b	int8	GPTQ-INT8 (W8A16)	85.12	59.42	1.43	w2_single_prefill
throughput	llamacpp	llama32-1b	16bit	F16 (GGUF)	448.95	448.95	1.00	w3_concurrent_8
throughput	llamacpp	llama32-1b	4bit	Q4_K_M (GGUF)	462.95	448.95	1.03	w3_concurrent_8
throughput	llamacpp	llama32-1b	8bit	Q8_0 (GGUF)	548.88	448.95	1.22	w3_concurrent_8
throughput	llamacpp	qwen3-0.6b	16bit	BF16 (GGUF)	615.11	615.11	1.00	w3_concurrent_8
throughput	llamacpp	qwen3-0.6b	4bit	Q4_K_M (GGUF)	534.34	615.11	0.87	w3_concurrent_8
throughput	llamacpp	qwen3-0.6b	8bit	Q8_0 (GGUF)	504.71	615.11	0.82	w3_concurrent_8
throughput	llamacpp	qwen3-1.7b	16bit	BF16 (GGUF)	233.15	233.15	1.00	w3_concurrent_8
throughput	llamacpp	qwen3-1.7b	4bit	Q4_K_M (GGUF)	388.60	233.15	1.67	w3_concurrent_8
throughput	llamacpp	qwen3-1.7b	8bit	Q8_0 (GGUF)	371.16	233.15	1.59	w3_concurrent_8
throughput	vllm	llama32-1b	16bit	BF16 (unquantized)	105.55	105.55	1.00	w3_concurrent_8
throughput	vllm	llama32-1b	8bit	FP8 dynamic (W8A8)	153.51	105.55	1.45	w3_concurrent_8
throughput	vllm	qwen3-0.6b	16bit	BF16 (unquantized)	1,024.5	1,024.5	1.00	w3_concurrent_8
throughput	vllm	qwen3-0.6b	8bit	FP8 (W8A8, e4m3)	1,302.3	1,024.5	1.27	w3_concurrent_8
throughput	vllm	qwen3-0.6b	int8	GPTQ-INT8 (W8A16)	1,266.3	1,024.5	1.24	w3_concurrent_8
throughput	vllm	qwen3-1.7b	16bit	BF16 (unquantized)	470.97	470.97	1.00	w3_concurrent_8
throughput	vllm	qwen3-1.7b	8bit	FP8 (W8A8, e4m3)	679.11	470.97	1.44	w3_concurrent_8
throughput	vllm	qwen3-1.7b	int8	GPTQ-INT8 (W8A16)	691.55	470.97	1.47	w3_concurrent_8
throughput	llamacpp	llama32-1b	16bit	F16 (GGUF)	453.32	453.32	1.00	w4_concurrent_32
throughput	llamacpp	llama32-1b	4bit	Q4_K_M (GGUF)	505.68	453.32	1.12	w4_concurrent_32
throughput	llamacpp	llama32-1b	8bit	Q8_0 (GGUF)	445.91	453.32	0.98	w4_concurrent_32
throughput	llamacpp	qwen3-0.6b	16bit	BF16 (GGUF)	462.63	462.63	1.00	w4_concurrent_32
throughput	llamacpp	qwen3-0.6b	4bit	Q4_K_M (GGUF)	516.16	462.63	1.12	w4_concurrent_32
throughput	llamacpp	qwen3-0.6b	8bit	Q8_0 (GGUF)	492.42	462.63	1.06	w4_concurrent_32

axis	backend	model	quant_level	quant_scheme	value	reference	ratio	workload
throughput	llamacpp	qwen3-1.7b	16bit	BF16 (GGUF)	267.80	267.80	1.00	w4_concurrent_32
throughput	llamacpp	qwen3-1.7b	4bit	Q4_K_M (GGUF)	384.77	267.80	1.44	w4_concurrent_32
throughput	llamacpp	qwen3-1.7b	8bit	Q8_0 (GGUF)	378.57	267.80	1.41	w4_concurrent_32
throughput	vllm	llama32-1b	16bit	BF16 (unquantized)	370.65	370.65	1.00	w4_concurrent_32
throughput	vllm	llama32-1b	8bit	FP8 dynamic (W8A8)	485.94	370.65	1.31	w4_concurrent_32
throughput	vllm	qwen3-0.6b	16bit	BF16 (unquantized)	2,788.7	2,788.7	1.00	w4_concurrent_32
throughput	vllm	qwen3-0.6b	8bit	FP8 (W8A8, e4m3)	3,222.3	2,788.7	1.16	w4_concurrent_32
throughput	vllm	qwen3-0.6b	int8	GPTQ-INT8 (W8A16)	2,937.3	2,788.7	1.05	w4_concurrent_32
throughput	vllm	qwen3-1.7b	16bit	BF16 (unquantized)	1,446.3	1,446.3	1.00	w4_concurrent_32
throughput	vllm	qwen3-1.7b	8bit	FP8 (W8A8, e4m3)	1,870.4	1,446.3	1.29	w4_concurrent_32
throughput	vllm	qwen3-1.7b	int8	GPTQ-INT8 (W8A16)	1,859.7	1,446.3	1.29	w4_concurrent_32

Lifecycle — time-to-ready and peak memory

backend	model	quant_level	quant_scheme	startup_s	gpu_peak_mib	host_rss_peak_mib
llamacpp	Qwen3-0.6B	4bit	Q4_K_M (GGUF)	2.06	4040	4,090.6
llamacpp	Qwen3-0.6B	8bit	Q8_0 (GGUF)	7.02	4417	4,190.8
llamacpp	Qwen3-0.6B	16bit	BF16 (GGUF)	3.01	5621	4,552.6
llamacpp	Qwen3-4B	4bit	Q4_K_M (GGUF)	14.07	6724	8,208.9
llamacpp	Qwen3-4B	8bit	Q8_0 (GGUF)	24.09	7876	1,602.9
llamacpp	Llama-3.2-1B-Instruct	4bit	Q4_K_M (GGUF)	5.13	3977	1,664.7
llamacpp	Llama-3.2-1B-Instruct	8bit	Q8_0 (GGUF)	7.02	4965	1,692.5
llamacpp	Llama-3.2-1B-Instruct	16bit	F16 (GGUF)	16.06	7301	2,125.7
llamacpp	Llama-3.2-3B-Instruct	4bit	Q4_K_M (GGUF)	11.07	7574	4,246.1
llamacpp	Llama-3.2-3B-Instruct	8bit	Q8_0 (GGUF)	33.41	7711	4,422.8
llamacpp	Qwen3-4B	4bit	Q4_K_M (GGUF)	4.08	7763	5,138.5
llamacpp	Qwen3-4B	8bit	Q8_0 (GGUF)	26.07	7766	5,271.4
vllm	Qwen3-0.6B	int8	GPTQ-INT8 (W8A16)	41.09	7883	3,810.8
vllm	Qwen3-0.6B	8bit	FP8 (W8A8, e4m3)	42.05	7874	3,817.7
vllm	Qwen3-0.6B	16bit	BF16 (unquantized)	41.06	7897	3,767.8
vllm	Qwen3-1.7B	int8	GPTQ-INT8 (W8A16)	45.06	7880	3,843.8
vllm	Qwen3-1.7B	8bit	FP8 (W8A8, e4m3)	73.30	7859	3,817.7
vllm	Qwen3-1.7B	16bit	BF16 (unquantized)	79.13	7823	3,764.3
vllm	Qwen3-4B	4bit	AWQ-INT4 (W4A16)	91.14	7907	3,932.6
vllm	Qwen3-4B	8bit	FP8 (W8A8, e4m3)	132.19	7920	3,865.8
vllm	Llama-3.2-1B-Instruct	8bit	FP8 dynamic (W8A8)	116.24	7839	4,044.7
vllm	Llama-3.2-1B-Instruct	16bit	BF16 (unquantized)	99.19	7730	3,956.5
vllm	Llama-3.2-3B-Instruct	8bit	FP8 dynamic (W8A8)	165.28	7595	4,179.5
llamacpp	Qwen3-1.7B	4bit	Q4_K_M (GGUF)	3.01	5210	4,162.5
llamacpp	Qwen3-1.7B	8bit	Q8_0 (GGUF)	8.02	6765	4,193.3
llamacpp	Qwen3-1.7B	16bit	BF16 (GGUF)	27.07	7593	4,960.9

backend	model	quant_level	quant_scheme	startup_s	gpu_peak_mib	host_rss_peak_mib
llamacpp	Qwen3-0.6B	4bit	Q4_K_M (GGUF)	2.01	2993	623.31
llamacpp	Qwen3-0.6B	8bit	Q8_0 (GGUF)	2.00	3225	641.17
vllm	Qwen3-0.6B	8bit	FP8 (W8A8, e4m3)	48.08	7863	3,678.1

Memory efficiency

`tok_s_per_mib` = output throughput per MiB of peak VRAM. `vram_mib_per_1b_params` is the footprint scaling factor.

workload	backend	model_key	model	params_b	quant_level	quant_scheme	out_tok_s	gpu_pe
w1_single_short	llamacpp	qwen3-0.6b	Qwen3-0.6B	0.60	4bit	Q4_K_M (GGUF)	238.84	2992
w1_single_short	llamacpp	llama32-1b	Llama-3.2-1B-Instruct	1.00	4bit	Q4_K_M (GGUF)	194.72	2827
w1_single_short	llamacpp	qwen3-0.6b	Qwen3-0.6B	0.60	8bit	Q8_0 (GGUF)	214.71	3481
w1_single_short	llamacpp	qwen3-1.7b	Qwen3-1.7B	1.70	4bit	Q4_K_M (GGUF)	146.11	3832
w1_single_short	llamacpp	llama32-1b	Llama-3.2-1B-Instruct	1.00	8bit	Q8_0 (GGUF)	133.71	3532
w1_single_short	llamacpp	qwen3-0.6b	Qwen3-0.6B	0.60	16bit	BF16 (GGUF)	134.69	3699
w1_single_short	llamacpp	qwen3-1.7b	Qwen3-1.7B	1.70	8bit	Q8_0 (GGUF)	108.67	4214
w1_single_short	vllm	qwen3-0.6b	Qwen3-0.6B	0.60	8bit	FP8 (W8A8, e4m3)	180.85	7240
w1_single_short	vllm	qwen3-0.6b	Qwen3-0.6B	0.60	int8	GPTQ-INT8 (W8A16)	183.51	7426
w1_single_short	llamacpp	llama32-1b	Llama-3.2-1B-Instruct	1.00	16bit	F16 (GGUF)	88.21	4637
w1_single_short	vllm	qwen3-0.6b	Qwen3-0.6B	0.60	16bit	BF16 (unquantized)	130.22	6934
w1_single_short	llamacpp	llama32-3b	Llama-3.2-3B-Instruct	3.00	4bit	Q4_K_M (GGUF)	98.22	5270
w1_single_short	vllm	qwen3-1.7b	Qwen3-1.7B	1.70	8bit	FP8 (W8A8, e4m3)	92.35	6970
w1_single_short	vllm	qwen3-1.7b	Qwen3-1.7B	1.70	int8	GPTQ-INT8 (W8A16)	92.19	7067
w1_single_short	llamacpp	qwen3-4b	Qwen3-4B	4.00	4bit	Q4_K_M (GGUF)	75.00	5916
w1_single_short	llamacpp	qwen3-1.7b	Qwen3-1.7B	1.70	16bit	BF16 (GGUF)	62.83	5834
w1_single_short	vllm	qwen3-4b	Qwen3-4B	4.00	4bit	AWQ-INT4 (W4A16)	71.55	7150
w1_single_short	llamacpp	llama32-3b	Llama-3.2-3B-Instruct	3.00	8bit	Q8_0 (GGUF)	66.89	6733
w1_single_short	vllm	qwen3-1.7b	Qwen3-1.7B	1.70	16bit	BF16 (unquantized)	63.23	6882
w1_single_short	vllm	qwen3-4b	Qwen3-4B	4.00	8bit	FP8 (W8A8, e4m3)	50.37	7415
w1_single_short	llamacpp	qwen3-4b	Qwen3-4B	4.00	8bit	Q8_0 (GGUF)	48.75	7539
w1_single_short	vllm	llama32-1b	Llama-3.2-1B-Instruct	1.00	8bit	FP8 dynamic (W8A8)	31.46	7218
w1_single_short	vllm	llama32-1b	Llama-3.2-1B-Instruct	1.00	16bit	BF16 (unquantized)	15.29	7331
w1_single_short	vllm	llama32-3b	Llama-3.2-3B-Instruct	3.00	8bit	FP8 dynamic (W8A8)	11.42	7296
w2_single_prefill	llamacpp	llama32-1b	Llama-3.2-1B-Instruct	1.00	4bit	Q4_K_M (GGUF)	185.74	2827
w2_single_prefill	llamacpp	qwen3-0.6b	Qwen3-0.6B	0.60	4bit	Q4_K_M (GGUF)	192.50	2992

workload	backend	model_key	model	params_b	quant_level	quant_scheme	out_tok_s	gpu_pe
w2_single_prefill	llamacpp	qwen3-0.6b	Qwen3-0.6B	0.60	8bit	Q8_0 (GGUF)	185.57	3481
w2_single_prefill	llamacpp	llama32-1b	Llama-3.2-1B-Instruct	1.00	8bit	Q8_0 (GGUF)	128.62	3532
w2_single_prefill	llamacpp	qwen3-1.7b	Qwen3-1.7B	1.70	4bit	Q4_K_M (GGUF)	125.30	3832
w2_single_prefill	llamacpp	qwen3-0.6b	Qwen3-0.6B	0.60	16bit	BF16 (GGUF)	119.04	3699
w2_single_prefill	vllm	qwen3-0.6b	Qwen3-0.6B	0.60	8bit	FP8 (W8A8, e4m3)	175.94	6904
w2_single_prefill	vllm	qwen3-0.6b	Qwen3-0.6B	0.60	int8	GPTQ-INT8 (W8A16)	172.33	7426
w2_single_prefill	llamacpp	qwen3-1.7b	Qwen3-1.7B	1.70	8bit	Q8_0 (GGUF)	90.74	4214
w2_single_prefill	llamacpp	llama32-1b	Llama-3.2-1B-Instruct	1.00	16bit	F16 (GGUF)	83.95	4637
w2_single_prefill	vllm	qwen3-0.6b	Qwen3-0.6B	0.60	16bit	BF16 (unquantized)	116.59	6960
w2_single_prefill	llamacpp	llama32-3b	Llama-3.2-3B-Instruct	3.00	4bit	Q4_K_M (GGUF)	86.38	5270
w2_single_prefill	vllm	qwen3-1.7b	Qwen3-1.7B	1.70	8bit	FP8 (W8A8, e4m3)	86.54	6970
w2_single_prefill	vllm	qwen3-1.7b	Qwen3-1.7B	1.70	int8	GPTQ-INT8 (W8A16)	85.12	7067
w2_single_prefill	llamacpp	qwen3-4b	Qwen3-4B	4.00	4bit	Q4_K_M (GGUF)	62.62	5951
w2_single_prefill	llamacpp	qwen3-1.7b	Qwen3-1.7B	1.70	16bit	BF16 (GGUF)	55.18	5841
w2_single_prefill	llamacpp	llama32-3b	Llama-3.2-3B-Instruct	3.00	8bit	Q8_0 (GGUF)	60.91	6733
w2_single_prefill	vllm	qwen3-4b	Qwen3-4B	4.00	4bit	AWQ-INT4 (W4A16)	62.87	7150
w2_single_prefill	vllm	qwen3-1.7b	Qwen3-1.7B	1.70	16bit	BF16 (unquantized)	59.42	6882
w2_single_prefill	vllm	qwen3-4b	Qwen3-4B	4.00	8bit	FP8 (W8A8, e4m3)	44.18	7415
w2_single_prefill	llamacpp	qwen3-4b	Qwen3-4B	4.00	8bit	Q8_0 (GGUF)	43.38	7539
w2_single_prefill	vllm	llama32-1b	Llama-3.2-1B-Instruct	1.00	8bit	FP8 dynamic (W8A8)	19.19	7218
w2_single_prefill	vllm	llama32-1b	Llama-3.2-1B-Instruct	1.00	16bit	BF16 (unquantized)	13.71	7331
w2_single_prefill	vllm	llama32-3b	Llama-3.2-3B-Instruct	3.00	8bit	FP8 dynamic (W8A8)	11.29	7296
w3_concurrent_8	vllm	qwen3-0.6b	Qwen3-0.6B	0.60	8bit	FP8 (W8A8, e4m3)	1,302.3	7076
w3_concurrent_8	llamacpp	qwen3-0.6b	Qwen3-0.6B	0.60	4bit	Q4_K_M (GGUF)	534.34	2992
w3_concurrent_8	vllm	qwen3-0.6b	Qwen3-0.6B	0.60	int8	GPTQ-INT8 (W8A16)	1,266.3	7426
w3_concurrent_8	llamacpp	qwen3-0.6b	Qwen3-0.6B	0.60	16bit	BF16 (GGUF)	615.11	3699

workload	backend	model_key	model	params_b	quant_level	quant_scheme	out_tok_s	gpu_pe
w3_concurrent_8	llamacpp	llama32-1b	Llama-3.2-1B-Instruct	1.00	8bit	Q8_0 (GGUF)	548.88	3532
w3_concurrent_8	llamacpp	llama32-1b	Llama-3.2-1B-Instruct	1.00	4bit	Q4_K_M (GGUF)	462.95	3019
w3_concurrent_8	vllm	qwen3-0.6b	Qwen3-0.6B	0.60	16bit	BF16 (unquantized)	1,024.5	6981
w3_concurrent_8	llamacpp	qwen3-0.6b	Qwen3-0.6B	0.60	8bit	Q8_0 (GGUF)	504.71	3481
w3_concurrent_8	llamacpp	qwen3-1.7b	Qwen3-1.7B	1.70	4bit	Q4_K_M (GGUF)	388.60	3832
w3_concurrent_8	vllm	qwen3-1.7b	Qwen3-1.7B	1.70	int8	GPTQ-INT8 (W8A16)	691.55	7067
w3_concurrent_8	vllm	qwen3-1.7b	Qwen3-1.7B	1.70	8bit	FP8 (W8A8, e4m3)	679.11	6970
w3_concurrent_8	llamacpp	llama32-1b	Llama-3.2-1B-Instruct	1.00	16bit	F16 (GGUF)	448.95	4637
w3_concurrent_8	llamacpp	qwen3-1.7b	Qwen3-1.7B	1.70	8bit	Q8_0 (GGUF)	371.16	4494
w3_concurrent_8	vllm	qwen3-4b	Qwen3-4B	4.00	4bit	AWQ-INT4 (W4A16)	509.56	7150
w3_concurrent_8	vllm	qwen3-1.7b	Qwen3-1.7B	1.70	16bit	BF16 (unquantized)	470.97	6882
w3_concurrent_8	vllm	qwen3-4b	Qwen3-4B	4.00	8bit	FP8 (W8A8, e4m3)	362.52	7415
w3_concurrent_8	llamacpp	qwen3-1.7b	Qwen3-1.7B	1.70	16bit	BF16 (GGUF)	233.15	5916
w3_concurrent_8	llamacpp	llama32-3b	Llama-3.2-3B-Instruct	3.00	8bit	Q8_0 (GGUF)	262.53	6733
w3_concurrent_8	llamacpp	llama32-3b	Llama-3.2-3B-Instruct	3.00	4bit	Q4_K_M (GGUF)	199.81	5270
w3_concurrent_8	llamacpp	qwen3-4b	Qwen3-4B	4.00	4bit	Q4_K_M (GGUF)	187.22	5951
w3_concurrent_8	llamacpp	qwen3-4b	Qwen3-4B	4.00	8bit	Q8_0 (GGUF)	188.36	7545
w3_concurrent_8	vllm	llama32-1b	Llama-3.2-1B-Instruct	1.00	8bit	FP8 dynamic (W8A8)	153.51	7218
w3_concurrent_8	vllm	llama32-3b	Llama-3.2-3B-Instruct	3.00	8bit	FP8 dynamic (W8A8)	107.49	7389
w3_concurrent_8	vllm	llama32-1b	Llama-3.2-1B-Instruct	1.00	16bit	BF16 (unquantized)	105.55	7331
w4_concurrent_32	vllm	qwen3-0.6b	Qwen3-0.6B	0.60	8bit	FP8 (W8A8, e4m3)	3,222.3	7076
w4_concurrent_32	vllm	qwen3-0.6b	Qwen3-0.6B	0.60	16bit	BF16 (unquantized)	2,788.7	6981
w4_concurrent_32	vllm	qwen3-0.6b	Qwen3-0.6B	0.60	int8	GPTQ-INT8 (W8A16)	2,937.3	7426
w4_concurrent_32	vllm	qwen3-1.7b	Qwen3-1.7B	1.70	8bit	FP8 (W8A8, e4m3)	1,870.4	6970
w4_concurrent_32	vllm	qwen3-1.7b	Qwen3-1.7B	1.70	int8		1,859.7	7067

workload	backend	model_key	model	params_b	quant_level	quant_scheme	out_tok_s	gpu_pe
						GPTQ-INT8 (W8A16)		
w4_concurrent_32	vllm	qwen3-1.7b	Qwen3-1.7B	1.70	16bit	BF16 (unquantized)	1,446.3	6882
w4_concurrent_32	vllm	qwen3-4b	Qwen3-4B	4.00	4bit	AWQ-INT4 (W4A16)	1,260.4	7150
w4_concurrent_32	llamacpp	qwen3-0.6b	Qwen3-0.6B	0.60	4bit	Q4_K_M (GGUF)	516.16	2994
w4_concurrent_32	llamacpp	llama32-1b	Llama-3.2-1B-Instruct	1.00	4bit	Q4_K_M (GGUF)	505.68	3019
w4_concurrent_32	llamacpp	qwen3-0.6b	Qwen3-0.6B	0.60	8bit	Q8_0 (GGUF)	492.42	3481
w4_concurrent_32	vllm	qwen3-4b	Qwen3-4B	4.00	8bit	FP8 (W8A8, e4m3)	1,031.5	7415
w4_concurrent_32	llamacpp	llama32-1b	Llama-3.2-1B-Instruct	1.00	8bit	Q8_0 (GGUF)	445.91	3532
w4_concurrent_32	llamacpp	qwen3-0.6b	Qwen3-0.6B	0.60	16bit	BF16 (GGUF)	462.63	3699
w4_concurrent_32	llamacpp	qwen3-1.7b	Qwen3-1.7B	1.70	4bit	Q4_K_M (GGUF)	384.77	3832
w4_concurrent_32	llamacpp	llama32-1b	Llama-3.2-1B-Instruct	1.00	16bit	F16 (GGUF)	453.32	4637
w4_concurrent_32	llamacpp	qwen3-1.7b	Qwen3-1.7B	1.70	8bit	Q8_0 (GGUF)	378.57	4507
w4_concurrent_32	vllm	llama32-1b	Llama-3.2-1B-Instruct	1.00	8bit	FP8 dynamic (W8A8)	485.94	7218
w4_concurrent_32	vllm	llama32-1b	Llama-3.2-1B-Instruct	1.00	16bit	BF16 (unquantized)	370.65	7331
w4_concurrent_32	llamacpp	qwen3-1.7b	Qwen3-1.7B	1.70	16bit	BF16 (GGUF)	267.80	5980
w4_concurrent_32	llamacpp	llama32-3b	Llama-3.2-3B-Instruct	3.00	4bit	Q4_K_M (GGUF)	224.56	5270
w4_concurrent_32	vllm	llama32-3b	Llama-3.2-3B-Instruct	3.00	8bit	FP8 dynamic (W8A8)	308.59	7389
w4_concurrent_32	llamacpp	llama32-3b	Llama-3.2-3B-Instruct	3.00	8bit	Q8_0 (GGUF)	269.55	6733
w4_concurrent_32	llamacpp	qwen3-4b	Qwen3-4B	4.00	4bit	Q4_K_M (GGUF)	186.38	5951
w4_concurrent_32	llamacpp	qwen3-4b	Qwen3-4B	4.00	8bit	Q8_0 (GGUF)	205.25	7568

Quality

`ppl_method` distinguishes llama.cpp's offline perplexity tool from vLLM's server-side echo path — they are different measurements and must not be compared directly.

backend	model	quant_level	quant_scheme	ppl	ppl_stderr	ppl_context_tokens	ppl_method	mc_ac
llamacpp	Qwen3-0.6B	4bit	Q4_K_M (GGUF)				server echo logprobs	
llamacpp	Qwen3-0.6B	8bit	Q8_0 (GGUF)				server echo logprobs	
llamacpp	Qwen3-0.6B	16bit	BF16 (GGUF)	15.25	0.22	2,048.0	llama.cpp perplexity tool (offline)	
llamacpp	Qwen3-4B	4bit	Q4_K_M (GGUF)	10.46	0.15	2,048.0	llama.cpp perplexity tool (offline)	
llamacpp	Qwen3-4B	8bit	Q8_0 (GGUF)	10.37	0.15	2,048.0	llama.cpp perplexity tool (offline)	
llamacpp	Llama-3.2-1B-Instruct	4bit	Q4_K_M (GGUF)	10.64	0.13	2,048.0	llama.cpp perplexity tool (offline)	
llamacpp	Llama-3.2-1B-Instruct	8bit	Q8_0 (GGUF)	10.41	0.13	2,048.0	llama.cpp perplexity tool (offline)	
llamacpp	Llama-3.2-1B-Instruct	16bit	F16 (GGUF)	10.42	0.13	2,048.0	llama.cpp perplexity tool (offline)	
llamacpp	Llama-3.2-3B-Instruct	4bit	Q4_K_M (GGUF)	8.24	0.10	2,048.0	llama.cpp perplexity tool (offline)	
llamacpp	Llama-3.2-3B-Instruct	8bit	Q8_0 (GGUF)	8.15	0.10	2,048.0	llama.cpp perplexity tool (offline)	
vllm	Qwen3-0.6B	int8	GPTQ-INT8 (W8A16)	15.00			server echo logprobs	0.26
vllm	Qwen3-0.6B	8bit	FP8 (W8A8, e4m3)	15.22			server echo logprobs	0.20
vllm	Qwen3-0.6B	16bit	BF16 (unquantized)	14.94			server echo logprobs	0.26
vllm	Qwen3-1.7B	int8	GPTQ-INT8 (W8A16)	12.52			server echo logprobs	0.23
vllm	Qwen3-1.7B	8bit	FP8 (W8A8, e4m3)	12.43			server echo logprobs	0.25
vllm	Qwen3-1.7B	16bit	BF16 (unquantized)	12.52			server echo logprobs	0.23
vllm	Qwen3-4B	4bit	AWQ-INT4 (W4A16)	11.41			server echo logprobs	0.25
vllm	Qwen3-4B	8bit	FP8 (W8A8, e4m3)	10.46			server echo logprobs	0.16
vllm	Llama-3.2-1B-Instruct	8bit	FP8 dynamic (W8A8)	10.74			server echo logprobs	0.30
vllm		16bit		10.67				0.30

backend	model	quant_level	quant_scheme	ppl	ppl_stderr	ppl_context_tokens	ppl_method	mc_ac
	Llama-3.2-1B-Instruct		BF16 (unquantized)				server echo logprobs	
vllm	Llama-3.2-3B-Instruct	8bit	FP8 dynamic (W8A8)	8.59			server echo logprobs	0.48
llamacpp	Qwen3-1.7B	4bit	Q4_K_M (GGUF)	13.37	0.21	2,048.0	llama.cpp perplexity tool (offline)	
llamacpp	Qwen3-1.7B	8bit	Q8_0 (GGUF)	13.00	0.20	2,048.0	llama.cpp perplexity tool (offline)	
llamacpp	Qwen3-1.7B	16bit	BF16 (GGUF)	13.04	0.20	2,048.0	llama.cpp perplexity tool (offline)	

Capability gaps

Recorded rather than hidden: these are properties of the backends, not failures of the harness.

category	backend	model	quant_level	detail
mc_unavailable	llamacpp	qwen3-0.6b	4bit	top-logprobs not served by llama.cpp /v1/completions
mc_unavailable	llamacpp	qwen3-0.6b	8bit	top-logprobs not served by llama.cpp /v1/completions
mc_unavailable	llamacpp	qwen3-0.6b	16bit	top-logprobs not served by llama.cpp /v1/completions
mc_unavailable	llamacpp	qwen3-4b	4bit	top-logprobs not served by llama.cpp /v1/completions
mc_unavailable	llamacpp	qwen3-4b	8bit	top-logprobs not served by llama.cpp /v1/completions
mc_unavailable	llamacpp	llama32-1b	4bit	top-logprobs not served by llama.cpp /v1/completions
mc_unavailable	llamacpp	llama32-1b	8bit	top-logprobs not served by llama.cpp /v1/completions
mc_unavailable	llamacpp	llama32-1b	16bit	top-logprobs not served by llama.cpp /v1/completions
mc_unavailable	llamacpp	llama32-3b	4bit	top-logprobs not served by llama.cpp /v1/completions
mc_unavailable	llamacpp	llama32-3b	8bit	top-logprobs not served by llama.cpp /v1/completions
mc_unavailable	llamacpp	qwen3-4b	4bit	top-logprobs not served by llama.cpp /v1/completions
mc_unavailable	llamacpp	qwen3-4b	8bit	top-logprobs not served by llama.cpp /v1/completions
startup_failure	vllm	qwen3-0.6b	int8	
startup_failure	vllm	qwen3-0.6b	8bit	
startup_failure	vllm	qwen3-0.6b	16bit	
startup_failure	vllm	qwen3-1.7b	int8	
startup_failure	vllm	qwen3-1.7b	8bit	
startup_failure	vllm	qwen3-1.7b	16bit	
startup_failure	vllm	qwen3-4b	4bit	
startup_failure	vllm	qwen3-4b	8bit	
startup_failure	vllm	llama32-1b	8bit	
startup_failure	vllm	llama32-1b	16bit	
startup_failure	vllm	llama32-3b	8bit	
startup_failure	vllm_gguf	qwen3-4b	4bit	
startup_failure	vllm_gguf	llama32-1b	4bit	
startup_failure	vllm_gguf	llama32-3b	4bit	
mc_unavailable	llamacpp	qwen3-1.7b	4bit	top-logprobs not served by llama.cpp /v1/completions
mc_unavailable	llamacpp	qwen3-1.7b	8bit	

category	backend	model	quant_level	detail
				top-logprobs not served by llama.cpp /v1/completions
mc_unavailable	llamacpp	qwen3-1.7b	16bit	top-logprobs not served by llama.cpp /v1/completions
perplexity_unavailable	llamacpp	qwen3-0.6b	4bit	error
mc_unavailable	llamacpp	qwen3-0.6b	4bit	top-logprobs not served by llama.cpp /v1/completions
perplexity_unavailable	llamacpp	qwen3-0.6b	8bit	error
mc_unavailable	llamacpp	qwen3-0.6b	8bit	top-logprobs not served by llama.cpp /v1/completions

Why there is no same-weights comparison

This was the intended anchor of the whole benchmark and it could not be produced.

Reason. vLLM 0.31 does not support GGUF. llama.cpp consumes GGUF (Q4_K_M/Q8_0/BF16); vLLM consumes safetensors with AWQ, FP8, GPTQ or compressed-tensors. The two quantization surfaces are disjoint, so 'same weights, different backend' cannot be measured by construction. Any llama.cpp-vs-vLLM throughput delta therefore confounds backend with quantization scheme and model artifact.

Verified by:

- no 'gguf' key in vllm QUANTIZATION_METHODS
- ModuleNotFoundError: vllm.model_executor.layers.quantization.gguf
- ImportError: cannot import GGUFConfig from vllm.transformers_utils.config
- serving a local .gguf path -> OSError 'not a valid JSON file'
- serving 'repo/file.gguf' -> OSError malformed repo id